✧ Fractions Made Fun ✧

Dividing Fractions

Keep • Change • Flip • Multiply • Simplify

1. What Is a Fraction?

A fraction represents a **part of a whole**. When something is divided into equal parts, each part is a fraction.

Term	Meaning	Example
Numerator	Top number – how many parts we have	3 in 3/4
Denominator	Bottom number – total equal parts	4 in 3/4
Proper fraction	Numerator < Denominator	2/5
Improper fraction	Numerator > Denominator	7/4
Mixed number	Whole number + fraction	1 and 3/4

2. What Does Dividing Fractions Mean?

Dividing fractions means finding out **how many times one fraction fits into another**.

Think of it like asking: *"How many halves fit into three quarters?"*

■ **Real-life Example**	You have **3/4** of a pizza. Each person gets **1/2** a pizza's worth. How many people can you feed? **Answer: 1 and 1/2 people!**

3. The Five Steps: Keep – Change – Flip – Multiply – Simplify

1	**KEEP**	Keep the first fraction exactly as it is.	**3/4 ÷ 1/2 → Keep 3/4**
2	**CHANGE**	Change the division sign (÷) to a multiplication sign (×).	**3/4 ÷ 1/2 → 3/4 × ...**

3	FLIP	Flip the second fraction (swap numerator and denominator). This is called finding the reciprocal.	**1/2 flipped → 2/1**
4	MULTIPLY	Multiply straight across: top × top, bottom × bottom.	**3/4 × 2/1 = 6/4 (3×2=6, 4×1=4)**
5	SIMPLIFY	Reduce to lowest terms by dividing by the greatest common factor (GCF).	**6/4 = 3/2 = 1 and 1/2**

4. Worked Example: 2/3 ÷ 4/5

Step	Action	Working	Result
1 – Keep	Keep the first fraction	—	2/3
2 – Change	Change ÷ to ×	2/3 ×	...
3 – Flip	Flip 4/5 to get its reciprocal	4/5 → 5/4	5/4
4 – Multiply	2/3 × 5/4 = (2×5) / (3×4)	10 / 12	10/12
5 – Simplify	Divide top & bottom by GCF (2)	10÷2 / 12÷2	5/6 ✓

✔ Final Answer: 2/3 ÷ 4/5 = 5/6

5. Word Problems

Read each problem carefully, then use the **Keep–Change–Flip** method to solve it.

■ Q1. Chocolate Bars	Priya has **3/4** of a chocolate bar. She wants to share it equally among 3 friends. How much does each friend get? Answer: ________________________________
■ Q2. Recipe Ingredients	Arjun needs **2/3** cup of flour. His measuring cup holds **1/6** cup. How many scoops does he need? Answer: ________________________________
■ Q3. Ribbon Pieces	Meera has **4/5** metre of ribbon. She cuts it into pieces that are **1/10** metre each. How many pieces can she make? Answer: ________________________________

6. Practice Problems

Solve these on your own using the steps you've learned!

Q1. 2/3 ÷ 1/4 = ____________	Q2. 5/6 ÷ 2/3 = ____________
Q3. 3/5 ÷ 1/2 = ____________	Q4. 4/7 ÷ 2/5 = ____________
Q5. 1/2 ÷ 3/4 = ____________	Q6. 7/8 ÷ 1/4 = ____________

7. Common Mistakes to Avoid

✗ **Forgetting to flip the second fraction**	✔ Always flip the SECOND fraction, not the first!
✗ **Multiplying the wrong fractions**	✔ After flipping, multiply top×top and bottom×bottom.
✗ **Not simplifying the final answer**	✔ Always check if you can reduce your fraction further.
✗ **Flipping the first fraction instead of the second**	✔ Only the divisor (second fraction) gets flipped.

8. Key Takeaways

KEEP	CHANGE	FLIP	MULTIPLY	SIMPLIFY
Keep the first fraction as-is	Change ÷ to ×	Flip the second fraction	Multiply top×top bottom×bottom	Simplify to lowest terms

Answer Key

Problem	Working	Answer
2/3 ÷ 1/4	2/3 × 4/1 = 8/3	2 and 2/3
5/6 ÷ 2/3	5/6 × 3/2 = 15/12	1 and 1/4
3/5 ÷ 1/2	3/5 × 2/1 = 6/5	1 and 1/5
4/7 ÷ 2/5	4/7 × 5/2 = 20/14	1 and 3/7
1/2 ÷ 3/4	1/2 × 4/3 = 4/6	2/3

Problem	Working	Answer
7/8 ÷ 1/4	7/8 × 4/1 = 28/8	3 and 1/2
Word Q1	3/4 ÷ 3 = 3/4 × 1/3 = 3/12	1/4
Word Q2	2/3 ÷ 1/6 = 2/3 × 6/1 = 12/3	4 scoops
Word Q3	4/5 ÷ 1/10 = 4/5 × 10/1 = 40/5	8 pieces

✥ You've got this! Keep practising and fractions will feel like second nature. ✥